

Response to Reviewer Feedback

Dear Editor and Reviewers,

Thank you for the detailed critique. I treated the review as a request for a full redesign. The manuscript no longer claims that sonification diagnoses pathogenicity or replaces standard bioinformatics. Instead, it tests a narrower and more defensible question: whether deterministic windowed sonification can represent mutation/anomaly patterns, especially order changes that composition summaries miss.

Major Revisions

- Replaced the earlier three-sequence design with 50 NCBI RefSeq *Xylella fastidiosa* assemblies, balanced across five host-associated groups.
- Corrected the biological framing of *X. fastidiosa* as a bacterial plant pathogen.
- Changed the study aim from broad genome comparison to reference-guided genomic anomaly representation.
- Added explicit preprocessing, sonification maps, baseline composition features, permutation testing, AUC, precision@4, bootstrap confidence intervals, and public-data ethics statement.
- Added a central result showing a sonification-specific usecase: codon-order shuffle anomalies were missed by composition features but detected by codon64 and amino-class sonification features.
- Added figures, full accession appendix, code/data availability, and a revised reference list with the requested sonification literature.

Remaining Scope Limits

The revised manuscript is careful not to claim disease diagnosis, virulence prediction, or replacement of alignment/variant-calling workflows. It presents sonification as an interpretable representation layer for reference-guided anomaly windows.